

MC10136

Universal Hexadecimal Counter

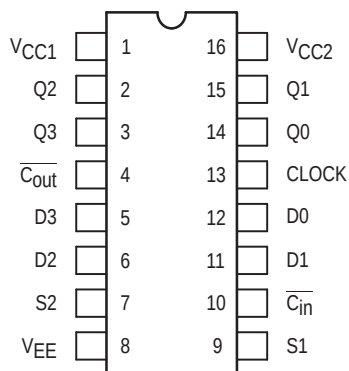
The MC10136 is a high speed synchronous counter that can count up, count down, preset, or stop count at frequencies exceeding 100 MHz. The flexibility of this device allows the designer to use one basic counter for most applications, and the synchronous count feature makes the MC10136 suitable for either computers or instrumentation.

Three control lines (S1, S2, and Carry In) determine the operation mode of the counter. Lines S1 and S2 determine one of four operations; preset (program), increment (count up), decrement (count down), or hold (stop count). Note that in the preset mode a clock pulse is necessary to load the counter, and the information present on the data inputs (D0, D1, D2, and D3) will be entered into the counter. Carry Out goes low on the terminal count, or when the counter is being preset.

This device is not designed for use with gated clocks. Control is via S1 and S2.

- $P_D = 625 \text{ mW typ/pkg (No Load)}$
- $f_{\text{count}} = 150 \text{ MHz typ}$
- $t_{pd} = 3.3 \text{ ns typ (C-Q)}$
- $7.0 \text{ ns typ (C-C}_{\text{out}})$
- $5.0 \text{ ns typ (C}_{\text{in}}\text{-C}_{\text{out}})$

DIP PIN ASSIGNMENT



Pin assignment is for Dual-in-Line Package.
For PLCC pin assignment, see the Pin Conversion Tables on page 18.

FUNCTION TABLE

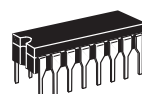
$\overline{C}_{in}$	S1	S2	Operating Mode
X	L	L	Preset (Program)
L	L	H	Increment (Count Up)
H	L	H	Hold Count
L	H	L	Decrement (Count Down)
H	H	L	Hold Count
X	H	H	Hold (Stop Count)



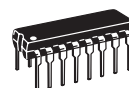
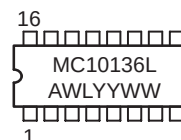
ON Semiconductor

<http://onsemi.com>

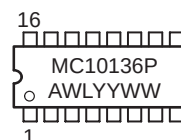
MARKING DIAGRAMS



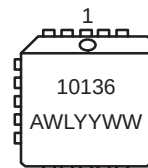
CDIP-16
L SUFFIX
CASE 620



PDIP-16
P SUFFIX
CASE 648



PLCC-20
FN SUFFIX
CASE 775



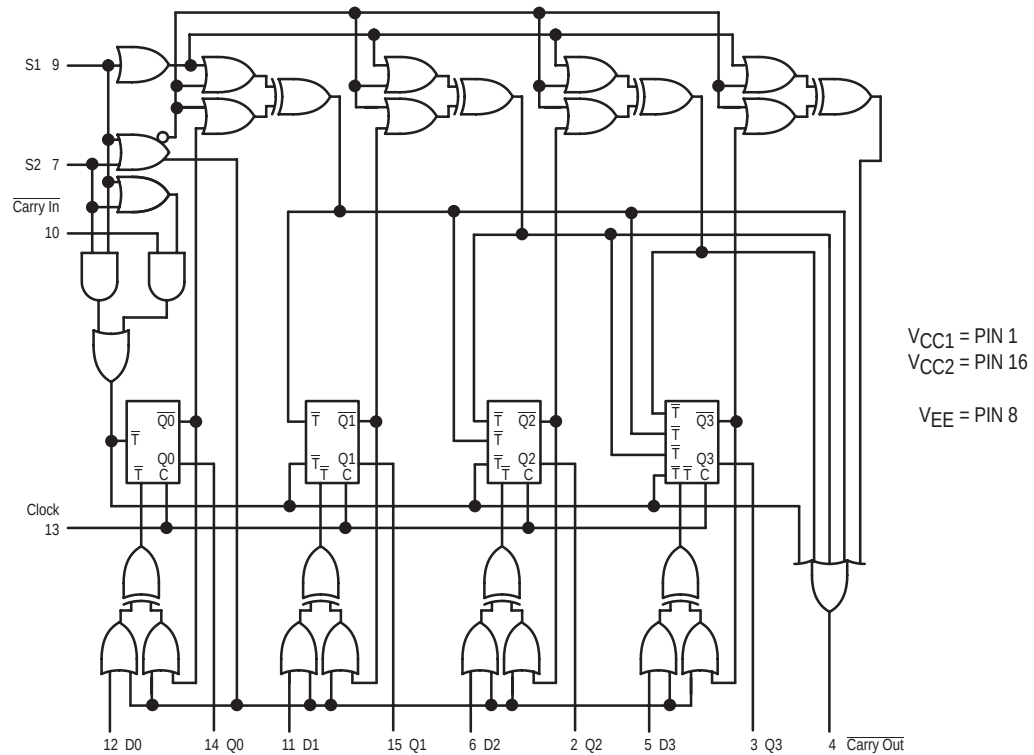
A = Assembly Location
WL = Wafer Lot
YY = Year
WW = Work Week

ORDERING INFORMATION

Device	Package	Shipping
MC10136L	CDIP-16	25 Units / Rail
MC10136P	PDIP-16	25 Units / Rail
MC10136FN	PLCC-20	46 Units / Rail

MC10136

LOGIC DIAGRAM



NOTE: Flip-flops will toggle when all $\bar{T}$ inputs are low.

SEQUENTIAL TRUTH TABLE*

INPUTS								OUTPUTS				
S1	S2	D0	D1	D2	D3	Carry In	Clock **	Q0	Q1	Q2	Q3	Carry Out
L	L	L	L	H	H	X	H	L	L	H	H	L
L	H	X	X	X	X	L	H	H	L	H	H	H
L	H	X	X	X	X	L	H	L	H	H	H	H
L	H	X	X	X	X	L	H	H	H	H	H	L
L	H	X	X	X	X	H	L	H	H	H	H	H
L	H	X	X	X	X	H	H	H	H	H	H	H
H	H	X	X	X	X	X	H	H	H	H	H	H
L	L	H	H	L	L	X	H	H	H	L	L	L
H	L	X	X	X	X	L	H	L	H	L	L	H
H	L	X	X	X	X	L	H	L	L	L	L	L
H	L	X	X	X	X	L	H	H	H	H	H	H

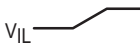
* Truth table shows logic states assuming inputs vary in sequence shown from top to bottom.

** A clock H is defined as a clock input transition from a low to a high logic level.

ELECTRICAL CHARACTERISTICS

Characteristic		Symbol	Pin Under Test	Test Limits						Unit	
				−30°C		+25°C			+85°C		
				Min	Max	Min	Typ	Max	Min		Max
Power Supply Drain Current		I _E	8		138		100	125		138	mAdc
Input Current		I _{inH}	5,6,11,12 7 9,10 13		350 425 390 460			220 265 245 290		220 265 245 290	μAdc
		I _{inL}	All	0.5		0.5			0.3		μAdc
Output Voltage	Logic 1	V _{OH}	14 (2.)	−1.060	−0.890	−0.960		−0.810	−0.890	−0.700	Vdc
Output Voltage	Logic 0	V _{OL}	14 (2.)	−1.890	−1.675	−1.850		−1.650	−1.825	−1.615	Vdc
Threshold Voltage	Logic 1	V _{OHA}	14 (2.)	−1.080		−0.980			−0.910		Vdc
Threshold Voltage	Logic 0	V _{OLA}	14 (2.)		−1.655			−1.630		−1.595	Vdc
Switching Times (50Ω Load)											ns
Propagation DelayClock Input		t ₁₃₊₁₄₊	14	0.8	4.8	1.0	3.3	4.5	1.4	5.0	
		t _{13+14−}	14	0.8	4.8	1.0	3.3	4.5	1.4	5.0	
		t ₁₃₊₄₊	4	2.0	10.9	2.5	7.0	10.5	2.4	11.5	
		t _{13+4−}	4	2.0	10.9	2.5	7.0	10.5	2.4	11.5	
$\overline{\text{Carry In}}$ to Carry Out		t _{10−4−}	4 (3.)	1.6	7.4	1.6	5.0	6.9	1.9	7.5	
		t ₁₀₊₄₊	4	1.6	7.4	1.6	5.0	6.9	1.9	7.5	
Setup Time	Data Inputs	t ₁₂₊₁₃₊	14	3.5		3.5			3.5		
		t _{12−13+}	14	3.5		3.5			3.5		
	Select Inputs	t ₉₊₁₃₊	14	6.0		6.0			6.0		
		t ₇₊₁₃₊	14	6.0		6.0			6.0		
$\overline{\text{Carry In}}$ Input		t _{10−13+}	14	2.5		2.5			3.0		
		t ₁₀₊₁₃₊	14	1.5		1.5			1.5		
Hold Time	Data Inputs	t ₁₃₊₁₂₊	14	0		0			0		
		t _{13+12−}	14	0		0			0		
	Select Inputs	t ₁₃₊₉₊	14	−1.0		−1.0			−1.0		
		t ₁₃₊₇₊	14	−1.0		−1.0			−1.0		
$\overline{\text{Carry In}}$ Input		t _{13+10−}	14	0		0			0		
		t ₁₃₊₁₀₊	14	0		0			0		
Counting Frequency		f _{countup}	14	125		125	150		125		MHz
		f _{countdown}	14	125		125	150		125		
Rise Time	(20 to 80%)	t ₄₊	4	0.9	3.3	1.1	2.0	3.3	1.1	3.5	ns
		t ₁₄₊	14	0.9	3.3	1.1	2.0	3.3	1.1	3.5	
Fall Time	(20 to 80%)	t _{4−}	4	0.9	3.3	1.1	2.0	3.3	1.1	3.5	
		t _{14−}	14	0.9	3.3	1.1	2.0	3.3	1.1	3.5	

1. Individually test each input; apply V_{ILmin} to pin under test.

2. Measure output after clock pulse  V_{IH} appears at clock input (Pin 13).

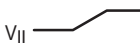
3. Before test set all Q outputs to a logic high.

4. To preserve reliable performance, the MC10136 (plastic packaged device only) is to be operated in ambient temperatures above 70°C only when 500lpm blown air or equivalent heat sinking is provided.

ELECTRICAL CHARACTERISTICS (continued)

@ Test Temperature			TEST VOLTAGE VALUES (Volts)					(V _{CC}) Gnd
			V _{IHmax}	V _{ILmin}	V _{IHAmin}	V _{ILAmax}	V _{EE}	
			–30°C	–0.890	–1.890	–1.205	–1.500	–5.2
			+25°C	–0.810	–1.850	–1.105	–1.475	–5.2
			+85°C	–0.700	–1.825	–1.035	–1.440	–5.2
Characteristic	Symbol	Pin Under Test	TEST VOLTAGE APPLIED TO PINS LISTED BELOW					(V _{CC}) Gnd
			V _{IHmax}	V _{ILmin}	V _{IHAmin}	V _{ILAmax}	V _{EE}	
Power Supply Drain Current	I _E	8					8	1, 16
Input Current	I _{inH}	5,6,11,12	5,6,11,12				8	1, 16
		7	7				8	1, 16
		9,10	9,10				8	1, 16
		13	13				8	1, 16
	I _{inL}	All		Note 1.			8	1, 16
Output Voltage Logic 1	V _{OH}	14 (2.)	12	7, 9			8	1, 16
Output Voltage Logic 0	V _{OL}	14 (2.)		7, 9			8	1, 16
Threshold Voltage Logic 1	V _{OHA}	14 (2.)		7, 9	12		8	1, 16
Threshold Voltage Logic 0	V _{OLA}	14 (2.)		7, 9		12	8	1, 16
Switching Times (50Ω Load)			+1.11V	+0.31V	Pulse In	Pulse Out	–3.2 V	+2.0 V
Propagation Delay Clock Input	t ₁₃₊₁₄₊	14	12		13	14	8	1, 16
	t _{13+14–}	14			13	14	8	1, 16
	t ₁₃₊₄₊	4	7		13	4	8	1, 16
	t _{13+4–}	4	7		13	4	8	1, 16
Carry In to Carry Out	t _{10–4–}	4 (3.)	7	13	10	4	8	1, 16
	t ₁₀₊₄₊	4	7	13	10	4	8	1, 16
Setup Time Data Inputs	t ₁₂₊₁₃₊	14		7, 9	12, 13	14	8	1, 16
	t _{12–13+}	14		7, 9	12, 13	14	8	1, 16
Select Inputs	t ₉₊₁₃₊	14			9, 13	14	8	1, 16
	t ₇₊₁₃₊	14			7, 13	14	8	1, 16
Carry In Inputs	t _{10–13+}	14	7	9	10, 13	14	8	1, 16
	t ₁₀₊₁₃₊	14	7	9	10, 13	14	8	1, 16
Hold Time Data Inputs	t ₁₃₊₁₂₊	14		7, 9	12, 13	14	8	1, 16
	t _{13+12–}	14		7, 9	12, 13	14	8	1, 16
Select Inputs	t ₁₃₊₉₊	14			9, 13	14	8	1, 16
	t ₁₃₊₇₊	14			7, 13	14	8	1, 16
Carry In Inputs	t _{13+10–}	14	7	9	10, 13	14	8	1, 16
	t ₁₃₊₁₀₊	14	7		10, 13	14	8	1, 16
Counting Frequency	f _{countup}	14	7		13	14	8	1, 16
	f _{countdown}	14	9		13	14	8	1, 16
Rise Time (20 to 80%)	t ₄₊	4	7		13	4	8	1, 16
	t ₁₄₊	14	7		13	14	8	1, 16
Fall Time (20 to 80%)	t _{4–}	4	7		13	4	8	1, 16
	t _{14–}	14	7		13	14	8	1, 16

1. Individually test each input; apply V_{ILmin} to pin under test.

2. Measure output after clock pulse  V_{IH} appears at clock input (Pin 13).

3. Before test set all Q outputs to a logic high.

4. To preserve reliable performance, the MC10136 (plastic packaged device only) is to be operated in ambient temperatures above 70°C only when 500lfpm blown air or equivalent heat sinking is provided.

Each MECL 10,000 series circuit has been designed to meet the dc specifications shown in the test table, after thermal equilibrium has been established. The circuit is in a test socket or mounted on a printed circuit board and transverse air flow greater than 500 linear fpm is maintained. Outputs are terminated through a 50-ohm resistor to –2.0 volts. Test procedures are shown for only one gate. The other gates are tested in the same manner.

MC10136

SWITCHING TIME TEST CIRCUIT AND WAVEFORMS @ 25°C

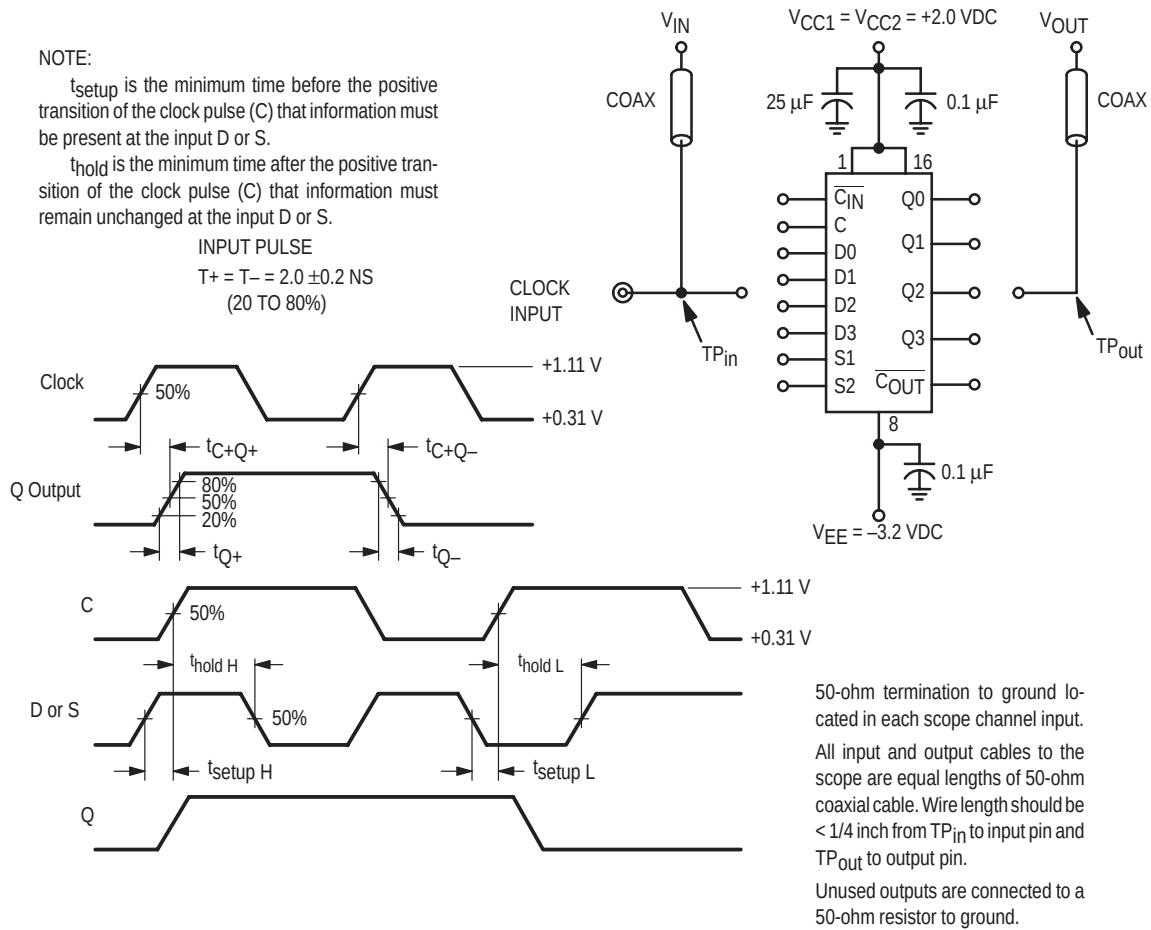
NOTE:

t_{setup} is the minimum time before the positive transition of the clock pulse (C) that information must be present at the input D or S.

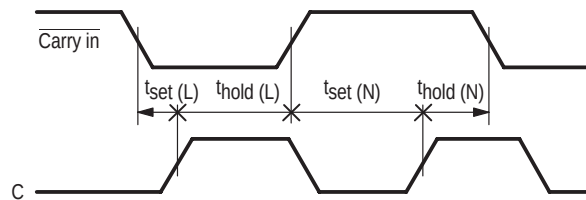
t_{hold} is the minimum time after the positive transition of the clock pulse (C) that information must remain unchanged at the input D or S.

INPUT PULSE

$$T+ = T- = 2.0 \pm 0.2 \text{ NS} \\ (20 \text{ TO } 80\%)$$



CARRY IN SET UP AND HOLD TIMES



APPLICATIONS INFORMATION

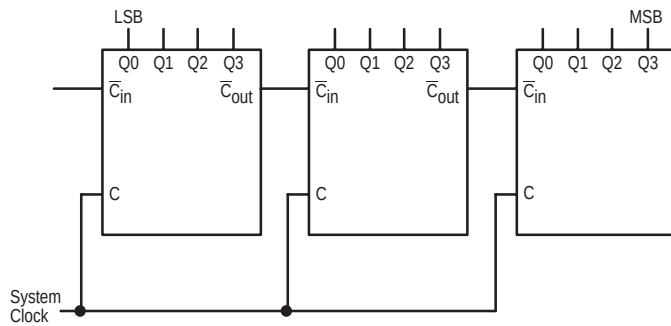
To provide more than four bits of counting capability several MC10136 counters may be cascaded. The Carry In input overrides the clock when the counter is either in the increment mode or the decrement mode of operation. This input allows several devices to be cascaded in a fully synchronous multistage counter as illustrated in Figure 1. The carry is advanced between stages as shown with no external gating. The Carry In of the first device may be left open. The system clock is common to all devices.

The various operational modes of the counter make it useful for a wide variety of applications. If used with MECL III devices, prescalers with input toggle frequencies in excess of 300 MHz are possible. Figure 2 shows such a prescaler using the MC10136 and MC1670. Use of the MC10231 in place of the MC1670 permits 200 MHz operation.

The MC10136 may also be used as a programmable counter. The configuration of Figure 3 requires no additional gates, although maximum frequency is limited to about 50 MHz. The divider modulus is equal to the program input plus one ($M = N + 1$), therefore, the counter will divide by a modulus varying from 1 to 16.

A second programmable configuration is also illustrated in Figure 4. A pulse swallowing technique is used to speed the counter operation up to 110 MHz typically. The divider modulus for this figure is equal to the program input ($M = N$). The minimum modulus is 2 because of the pulse swallowing technique, and the modulus may vary from 2 to 15. This programmable configuration requires an additional gate, such as $1/2$ MC10109 and a flip-flop such as $1/2$ MC10131.

Figure 1. 12 BIT SYNCHRONOUS COUNTER



NOTE: S1 and S2 are set either for increment or decrement operation.

Figure 2. 300 MHz PRESCALER

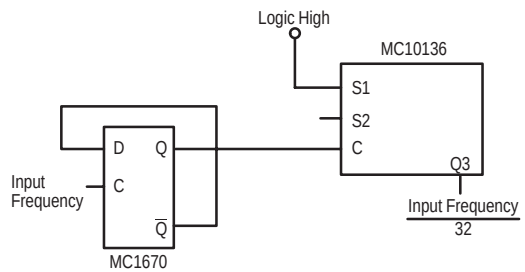
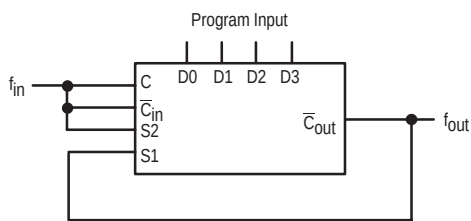
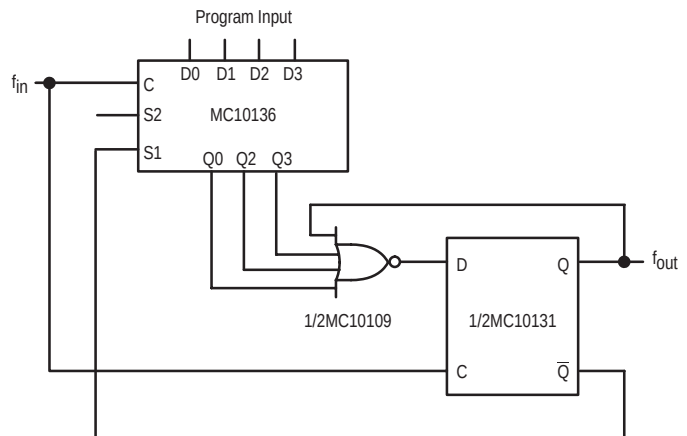


Figure 3. 50 MHz PROGRAMMABLE COUNTER



- 1 $f_{out} = \frac{f_{in}}{\text{Program Input} + 1}$
- 2 $f_{max} \approx 50 \text{ MHz Typ.}$
- 3 Divide Ratio is from 1 to 16.

Figure 4. 100 MHz PROGRAMMABLE COUNTER



- 1 $f_{out} = \frac{f_{in}}{\text{Program Input}}$
- 2 $f_{max} \approx 110 \text{ MHz Typ.}$
- 3 Divide Ratio is from 2 to 15.